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webofscience: <https://www.webofscience.com/wos/author/rid/N-6304-2019>

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Research Interest

Topics:

- Quantum machine learning and big data analysis
- Optimization methods on quantum and classical computers
- Quantum information, entanglement, algorithms, complexity, and circuits
- Adiabatic quantum computation and combinatorial optimization
- Numerical linear algebra and parallel and distributed computing
- Quantum bioinformatics

Education

- 2011 – 2014 **Ph.D., Dept. of Computer Science, Purdue University, West Lafayette, IN, US.**
Dissertation: *Quantum Circuit Design Methods and Applications, Purdue University ProQuest Dissertations & Theses, 2014. 3635993.*
Advisors: Prof. Ananth Grama and Prof. Sabre Kais
- 2009 – 2011 **M.Sc., Dept. of Computer Science, Purdue University, West Lafayette, IN, US.**
- 2008 – 2009 **ESL Student, ELS Language Center, New York, NY, US.**
EPI, University of South Carolina, Columbia, SC, US.
- 2002 – 2007 **B.Sc., Computer Engineering, Erciyes University, Kayseri, Turkiye.**

Employment History

- 2015 – **(May 2025- present) Associate Professor, (March 2015 - May 2025) Assistant Professor,** Department of Computer Engineering, Istanbul Medeniyet University, Istanbul, Turkey
served as department head, from January 2021 - February 2023 and vice department head from 2018 - 2021 and 2023-present.
- Summer 2013 **Research Consultant,** The Qatar Environment and Energy Research Institute (QEERI), QATAR.
- 2011 – 2012 **Research Assistant,** NSF Quantum Information for Quantum Chemistry Center, Purdue University, West Lafayette, Indiana, US.
- 2011 – 2013 **Teaching Assistant** Dept. of Computer Sciences, Purdue University, West Lafayette, Indiana, US.

Technical Strengths

Programming languages and technologies: Julia (currently experiencing), Rust(currently experiencing), Python (experience in numpy, pandas, scikit-learn, tensorflow, Qiskit), Java, C/C++, Matlab, MPI, OpenMP, SQL, MS Office, VBA, Linux, macOS, Windows.

Research Grants

- Application of Quantum Computation to the Problems in Bioinformatics and Chemistry.30000TL. Tubitak#115E747. (2016-2017)

Professional Activities

Co-organizer: Special Sessions on Quantum Cybernetics and Machine Learning at IEEE SMC 2019 and IEEE SMC 2020

Panellist: TUBITAK

Referee: Scientific Reports, Science Bulletin, IEEE Transactions on Pattern Analysis and Machine Intelligence, IEEE Transactions on Neural Networks and Learning Systems, Quantum Information and Processing, Quantum Machine Intelligence, IEEE Internet Computing, ACM Computing Surveys, IEEE Computer Architecture Letters, IEEE Transactions on Circuits and Systems I, Journal of Experimental and Theoretical Artificial Intelligence, Turkish Journal of Electrical Engineering and Computer Sciences, IEEE Transactions on Control Systems Technology, IEEE SMC Magazine

Memberships

IEEE Computer Society, IEEE SMC Quantum Cybernetics, ACM

Teaching

Experience

- **Istanbul Medeniyet University (with links to lecture notes):**
 - [Introduction to Quantum Computing](#) *Spring 2026*
 - [Operating Systems \(in Linux\)](#) *Fall 2020, 21, 21, 23, 24,25, Spring 26*
 - [System Programming \(in Unix\)](#) *Spring 2020, 21, 22, 23, 24, 25*
 - [Compiler Design](#) *Fall 2025*
 - [Data Mining](#) *Fall 2021*
 - [Data Structures and Algorithms \(in Java, C++, Python\)](#) *Fall 2019-2021*
 - [Discrete Mathematics](#) *Spring 2019, 2022, 2025*
 - [Introduction to Computer Science and Engineering](#) *Fall 2020, 2021, 2025*
 - **Computer Programming Courses in C/C++, Python, Java, MATLAB** *Fall 2016, 2017, 2018, 2019, 2021*
 - BIL491/492 Student Final Projects
 - BIL 206 Algorithm Analysis and Design *Spring 2022*
 - MY510 and MY610 Advanced Table Applications in Engineering(VBA), *Spring 2016, 2017*
- **Purdue University (Graduate Teaching Assistant):**
 - CS-18000 Problem Solving and Object Oriented Programming, *Fall 2011, 2012, Spring 2013*
 - CS-38100 Introduction to Algorithms *Fall 2012*

Publications

Journal Articles

1. Daskin, A. Quantum distortion model for running variational quantum algorithms without error corrections. *Discov. Quantum Sci.* 2, 2 (2026). <https://doi.org/10.1007/s44464-026-00006-y>
Keywords— quantum distortion, quantum error correction, quantum error mitigation
2. Daskin, A. Error analysis of quantum operators written as a linear combination of permutations. *Quantum Inf Process* 24, 149 (2025). <https://doi.org/10.1007/s11128-025-04771-0>
Keywords— quantum bit phase flip errors, error resilient algorithms, linear combination of permutations
3. Ammar Daskin, A unifying primary framework for quantum graph neural networks from quantum graph states, *Eur. Phys. J. Spec. Top.* (2024). <https://doi.org/10.1140/epjs/s11734-024-01382-1>
Keywords— quantum machine learning, graph neural networks, graph states
4. A Daskin, R Gupta, S Kais Dimension reduction and redundancy removal through successive Schmidt decompositions, *Appl. Sci.* 2023, 13(5), 3172; <https://doi.org/10.3390/app13053172>
Keywords— quantum machine learning, quantum algorithms, tensor decomposition, data mapping, dimension reduction
5. Ammar Daskin, Combinatorial optimization through variational quantum power method, *Quantum Inf Process* 20, 336 (2021). <https://doi.org/10.1007/s11128-021-03283-x>, [arXiv:2007.01004](https://arxiv.org/abs/2007.01004)
Keywords— quantum optimization, combinatorial optimization, power iteration, variational eigenvalue solver
6. Ammar Daskin, The quantum version of the shifted power method and its application in quadratic binary optimization, *Turk J Elec Eng and Comp Sci*, 2020, <https://doi.org/10.3906/elk-1910-99>, [arXiv:1809.01378](https://arxiv.org/abs/1809.01378)
Keywords— quantum optimization, power iteration, matrix computation, variational circuits
7. Ammar Daskin, Quantum Spectral Clustering through a Biased Phase Estimation Algorithm, *TWMS J. App. Eng. Math.* V.10, N.1, 2020, pp. 24-33, [arXiv:1703.05568](https://arxiv.org/abs/1703.05568)
Keywords— quantum machine learning, clustering, phase estimation, quantum amplitude amplification, principal components
8. Ammar Daskin, Teng Bian, Rongxin Xia, Sabre Kais, Context aware quantum simulation of a matrix stored in quantum memory, *Quantum Information Processing*, 18:357, 2019. <https://doi.org/10.1007/s11128-019-2469-1>
Keywords— quantum circuit design, quantum RAM, context aware circuits, fixed design circuits
9. Teng Bian, Daniel Murphy, Rongxin Xia, Ammar Daskin, Sabre Kais, Quantum computing methods for electronic states of the water molecule, *Molecular Physics*, 2019. <https://doi.org/10.1080/00268976.2019.1580392>
Keywords— quantum circuit design, quantum phase estimation, simulation of Water molecule with different approaches
10. Ammar Daskin, Sabre Kais, A Generalized Circuit for the Hamiltonian Dynamics Through the Truncated Series, *Quantum Information and Processing*, 17:328, 2018. <https://doi.org/10.1007/s11128-018-2099-z>
Keywords— quantum circuit design, non-unitary matrices, writing a matrix as a sum of unitaries, Taylor series
11. Ammar Daskin, A Quantum Implementation Model for Artificial Neural Networks, *Quanta*, 7:718, 2018. <http://dx.doi.org/10.12743/quanta.v7i1.65>
Keywords— quantum machine learning, Widrow-Hoff learning rule, quantum amplitude amplification, phase estimation, principal components
12. Ammar Daskin, Sabre Kais, Direct Application of the Phase Estimation Algorithm to Find the Eigenvalues of the Hamiltonians, *Chemical Physics*, Volume 514, Pages 87-94, 2018. <https://doi.org/10.1016/j.chemphys.2018.01.002>
Keywords— quantum circuit design, phase estimation, non-unitary matrices, sum of unitaries, polar form, plane rotations
13. Ammar Daskin, Sabre Kais, An Ancilla Based Quantum Simulation Framework for Non-Unitary Matrices, *Quantum Information and Processing*, vol.16, no.1, pp.33-43, 2017.
Keywords— quantum circuit design, phase estimation, non-unitary matrices, sum of unitaries, square root of a matrix, cosine-sine decomposition

14. Anmer Daskin, Obtaining A Linear Combination of the Principal Components of a Matrix on Quantum Computers, Quantum Information and Processing, Volume 15, Issue 10, pp 4013–4027, 2016. <http://dx.doi.org/10.1007/s11128-016-1388-7>
Keywords— quantum machine learning, quantum phase estimation, quantum amplitude amplification, principal component analysis
15. Anmer Daskin, Quantum Eigenvalue Estimation for Irreducible Non-negative Matrices, International Journal of Quantum Information, Volume 14, Issue 01, 2016. <http://dx.doi.org/10.1142/S0219749916500052>
Keywords— quantum phase estimation, nonnegative matrices, eigenvalues, phase estimation without state preparation
16. Katherine L. Brown, Anmer Daskin, Sabre Kais, Jonathan P. Dowling, Reducing the number of ancilla qubits and the gate count required for creating large controlled operations, Quantum Information and Processing, Volume 14, Issue 3, pp 891–899, 2015. <http://dx.doi.org/10.1007/s11128-014-0900-1>
Keywords— quantum circuit design, toffoli gates
17. Anmer Daskin, Ananth Grama, Sabre Kais, Quantum Random State Generation with Predefined Entanglement, 2014 Journal of Quantum Information, Vol. 12, No.5, 2014.
Keywords— quantum state preparation, entanglement, parameterized circuits, Schmidt decomposition
18. Anmer Daskin, Ananth Grama, Sabre Kais, Multiple Network Alignment on Quantum Computers, Quantum Information and Processing, Volume 13, Issue 12, pp 2653–2666, 2014. <http://dx.doi.org/10.1007/s11128-014-0818-7>
Keywords— quantum bioinformatics, quantum phase estimation, protein-protein interaction networks
19. Anmer Daskin, Ananth Grama, and Sabre Kais, A universal quantum circuit scheme for finding complex eigenvalues, Quantum Information Processing, Volume 13, Issue 2, pp 333–353, 2014, <http://dx.doi.org/10.1007/s11128-013-0654-1>
Keywords— quantum circuit design, non-unitary matrices, phase estimation, complex eigenvalues
20. Anmer Daskin, Ananth Grama, George Killios, and Sabre Kais, Universal Programmable Quantum Circuit Schemes to Emulate an Operator, J. Chem. Phys. 137, 234112, 2012. <http://dx.doi.org/10.1063/1.4772185>
Keywords— quantum circuit design in linear time, parameterized quantum circuits, sum of unitaries, Schmidt decomposition
21. Yudong Cao, Anmer Daskin, Steven Frankel, Sabre Kais, Quantum Circuit Design for Solving Linear Systems of Equations, Mol. Phys., 110, no. 15–16: 1675–1680, 2012. <http://dx.doi.org/10.1080/00268976.2012.668289>
Keywords— quantum circuit design, HHL algorithm, quantum phase estimation, matrix inverse
22. Anmer Daskin and Sabre Kais, Decomposition of Unitary Matrices for Finding Quantum Circuits: Application to Molecular Hamiltonians, J. Chem. Phys. 134, 144112, 2011. <http://dx.doi.org/10.1063/1.3575402>
Keywords— global optimization, quantum circuit design, quantum gates, quantum Hamiltonians, Hydrogen molecule, Water molecule
23. Anmer Daskin and Sabre Kais, Group Leaders Optimization Algorithm, Mol. Phys., 109(5):761–772, 2011. <http://dx.doi.org/10.1080/00268976.2011.552444>
Keywords— global optimization, Lennard-Jones potential, circuit optimization

Conference Publications

24. Daskin, Ammar. "Quantum RNNs and LSTMs Through Entangling and Disentangling Power of Unitary Transformations." In Proceedings of the 18th International Conference on Agents and Artificial Intelligence - Volume 2, ISBN 978-989-758-796-2, ISSN 2184-433X, pages 1076–1084, 2026. arXiv preprint [arXiv:2505.06774](https://arxiv.org/abs/2505.06774).
Keywords— quantum machine learning, time series analysis, LSTM, RNN
25. Ammar Daskin, A Simple Quantum Neural Net with a Periodic Activation Function, The 2018 IEEE International Conference on Systems, Man, and Cybernetics (SMC2018), Miyazaki, Japan. <https://doi.org/10.1109/SMC.2018.00491>
Keywords— quantum machine learning, quantum neural networks, periodic activation function, pattern recognition

26. Ammar Daskin , Quantum IsoRank: Efficient Alignment of Multiple Protein-Protein Interaction Networks, International Conference on Computer Science and Engineering (UBMK 2016), TEKIRDAG, TURKIYE, 20-23 Ekim 2016, pp.81-88 ([arXiv:1506.05905](#))
Keywords— quantum bioinformatics, quantum state tomography, protein-protein interaction networks

Conference Abstracts

27. Anmer Daskin, Quantum Circuit Implementations of the Direct Sum of Unitary Matrices, International Conference on Quantum Science and Applications, May 25-27, 2016, Eskisehir, Turkey
28. Anmer Daskin and Sabre Kais, Decomposition of Unitary Matrices for Finding Quantum Circuits, 241st ACS National Meeting & Exposition, The Division of Physical Chemistry, March 30, 2011, Anaheim, California, USA
29. Anmer Daskin, Finding Quantum Circuit Designs for the Simulation of Quantum Chemistry, 5th Annual Computational Science and Engineering Student Conference, 2013, Purdue University, W. Lafayette, USA

Preprints

30. Ammar Daskin, "Certifying Quantum Optimization and Circuit Cutting by Using Quantum-Classical Moment Duality," [arXiv:2606.23727](#), 2026. (Submitted)
Keywords— quantum optimization, circuit cutting, Sum-of-Squares duality, GW relaxation
31. Ammar Daskin, "Lowering LCU Circuit Width through Maximum-Weight Birkhoff-von Neumann Decomposition," [arXiv:2605.27430](#), 2026. (Submitted)
Keywords— quantum circuits, quantum compilers, linear combination of unitaries, Birkhoff-von Neumann, doubly stochastic matrices, circuit width
32. Ammar Daskin, "Exploiting all ancilla outcomes in linear combinations of unitaries: low-rank recovery and quantum trapdoor functions," [arXiv:2605.02986](#), 2026. (Submitted)
Keywords— quantum circuits, quantum tomography, linear combination of unitaries, low-rank approximation, quantum cryptography, trapdoor functions
33. Ammar Daskin, "Mitigating Frequency Learning Bias in Quantum Models via Multi-Stage Residual Learning," [arXiv:2603.10083](#), 2026.
Keywords— quantum machine learning, frequency bias, residual learning, Fourier parameterization
34. Ali Emre Aydin and Ammar Daskin, "Quantum Voting Protocol for Centralized and Distributed Voting Based on Phase-Flip Counting," [arXiv:2510.15243](#), 2025. (Submitted)
Keywords— quantum voting, quantum cryptography, phase-flip counting, entanglement
35. Ammar Daskin, "Dimension reduction with structure-aware quantum circuits for hybrid machine learning," [arXiv:2508.00048](#), 2025.
Keywords— quantum machine learning, quantum dimension reduction, SVD, Schmidt decomposition, structure-aware circuits
36. Ammar Daskin, "Learnable quantum spectral filters for hybrid graph neural networks," [arXiv:2507.05640](#), 2025.
Keywords— graph neural networks, Laplacian operator, quantum spectral filters, quantum Fourier circuits
37. Daskin, Ammar. "From Theory to Practice: Analyzing VQPM for Quantum Optimization of QUBO Problems." arXiv preprint [arXiv:2505.12990](#) (2025).
Keywords— Quantum optimization, variational quantum power method, qubo
38. Daskin, Ammar. "Quantum Simulations Based on Parameterized Circuit of an Antisymmetric Matrix." arXiv preprint [arXiv:2505.01023](#) (2025).
Keywords— Eigenvalue estimation, circuit approximation, quantum simulation
39. Daskin, Ammar. "An alternative explicit circuit diagram for the quantum search algorithm by implementing a non-unitary gate." arXiv preprint [arXiv:2412.16514](#) (2024).
Keywords— quantum search algorithm, grover search algorithm, unitary circuit

40. Ammar Daskin, A quantum compiler design method by using linear combinations of permutations, [arXiv:2404.18226](#), 2024
Keywords— quantum compiler, linear combination of permutations, Birkhoff-von Neumann algorithm, bistochastic matrices, quantum block encoding
41. Daskin, Ammar. "Quantum Kolmogorov-Arnold networks by combining quantum signal processing circuits." arXiv preprint [arXiv:2410.04218](#) (2024).
Keywords— Quantum Kolmogorov-Arnold networks, quantum KAN, quantum machine learning
42. Ammar Daskin, Federated learning with distributed fixed design quantum chips and quantum channels, [arXiv:2401.13421](#), 2024
Keywords— Federated learning, distributed quantum computing, machine learning, programmable chips
43. Ammar Daskin, A Simple Quantum Blockmodeling with Qubits and Permutations, [arXiv:2311.07726](#), 2023
Keywords— Blockmodeling, quantum optimization, qubit encoding, permutation matrices, Barbell graph
44. Ammar Daskin, On the explainability of quantum neural networks based on variational quantum circuits, [arXiv:2301.05549](#), 2023
Keywords— Quantum neural networks, explainability, interpretability, ridge functions
45. Ammar Daskin, Quantum implementation of circulant matrices and its use in quantum string processing, arXiv preprint [arXiv:2206.09364](#), 2022
Keywords— quantum machine learning, quantum algorithms, bioinformatics, circulant matrix, suffix trees, Burrows-Wheeler Transform, string processing
46. Ammar Daskin, A walk through of time series analysis on quantum computers, arXiv preprint [arXiv:2205.00986](#), 2022
Keywords— quantum machine learning, quantum optimization, forecasting models, autoregressive models, time series analysis
47. Ammar Daskin, A Quantum Approach to Subset-Sum and Similar Problems, arXiv preprint [arXiv:1707.08730](#), 2017
Keywords— quantum algorithms, combinatorial optimization, NP-hard problems, subset-sum problem, 0-1 knapsack problem, quantum Arthur-Merlin games